Session 4 Pre-Work

Skills, MCPs, and the Collaborative Story

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Session 4 introduces **MCPs** — the way Claude Code talks to systems outside your project. We will use two of them live: **GitHub** (for a collaborative writing exercise) and **Zotero** (for a citations demo). Both need to be installed and authenticated before class — they take 15–20 minutes each and the setup is too fiddly to do in the room.

Please complete **all six steps** below. Budget **45–60 minutes total**. If you get stuck for more than 10 minutes, stop and email me — don't try to debug it yourself. Once you are done, run the verification commands at the end and **send me a screenshot** of the output *and* your GitHub username (Step 2).

What You Need Before Starting

- Everything from the original Pre-Workshop Setup Guide working (Claude Code authenticated, VS Code running, Python installed)
- A GitHub account (created during initial setup)
- A free Zotero account — we'll create one in Step 5 if you don't have one

If `claude -version` or `git -version` doesn't work, fix that first — these steps assume the original setup is intact.

Step 1: Refresh the Course Repo and Set Up Your Workspace

Starting with Session 4 we are switching to a new workflow: the course repo stays **read-only** on your side, and you do all your actual work in a separate `~/my_workspace/` folder. This means I can push new material any time without breaking whatever you have in progress, and you can edit/break/delete freely without worrying about merge conflicts against the shared repo.

Run these three commands now, and plan to run them at the start of every future session too:

```
# 1.  Pull the latest course repo
cd ~/claude-code-workshop && git pull

# 2.  Copy today's session folder into your workspace
mkdir -p ~/my_workspace && cp -r session_4/ ~/my_workspace/session_4/

# 3.  Open your copy in VS Code and work there
cd ~/my_workspace/session_4 && code .
```

If you cloned the repo somewhere other than `~/claude-code-workshop` (e.g. inside Dropbox or Documents), substitute your actual path in step 1. If you don't already have the course repo cloned

at all, run this first:

```
cd ~ && git clone https://github.com/bradyallardice/claude-code-workshop.git
```

After Step 2, you should see a `session_4/` folder inside `~/my_workspace/` containing the new material. After Step 3, VS Code should open in that copy. From now on, **edit files in `~/my_workspace/session_4/`, not in the course repo itself.**

If `git pull` fails with a merge conflict, that is exactly the problem we fix in class — don't try to resolve it. Just email me and I will help you reset cleanly.

Step 2: Send Me Your GitHub Username

In Session 4 we run a **collaborative writing exercise** using a separate class GitHub repository. I need to add each of you as a collaborator before class so you can push branches and open pull requests.

Email me your GitHub username (not your email — the username, e.g. `octocat`) at brady.allardice@upf.edu. You can find it in the top-right of **https://github.com** after logging in.

Step 3: Update Claude Code

The MCP install commands below assume Claude Code 2.1 or newer. Update to the latest version:

```
claude update
```

Then confirm the version:

```
claude --version
```

Expected: 2.1.x or higher.

Step 4: Install the GitHub MCP

The GitHub MCP lets Claude Code open pull requests, leave review comments, and create issues directly from the chat. We use it during the collaborative story exercise.

4a. Create a GitHub Personal Access Token

A Personal Access Token (PAT) is a password-like string that lets the MCP act on your behalf. **Treat it like a password** — don't paste it into chat, don't commit it to git.

- Go to **https://github.com/settings/tokens?type=beta**
- Click **Generate new token**
- **Token name:** `claude-code-workshop`
- **Expiration:** 90 days
- **Repository access:** "All repositories" is fine for the workshop
- **Repository permissions:** grant **Read and write** on *Contents*, *Issues*, and *Pull requests*

- Click **Generate token**, then **copy the token immediately** — GitHub will not show it again

Paste it into a temporary scratch file you will delete after Step 4b.

4b. Add the GitHub MCP to Claude Code

In your terminal, paste the command below, **but before pressing Enter, replace the literal text `YOUR_PAT` with the token you just copied** (keep the word `Bearer` and the space before the token — you are only replacing `YOUR_PAT`, nothing else):

```
claude mcp add --scope user --transport http github https://api.githubcopilot.com/mcp/
-H "Authorization:  Bearer YOUR_PAT"
```

Example of what it should look like after substitution (with a fake token):

```
claude mcp add --scope user --transport http github https://api.githubcopilot.com/mcp/
-H "Authorization:  Bearer github_pat_11ABCDEFG0abc123XYZ..."
```

Do not drop `--scope user`. By default, `claude mcp add` registers a server only for the folder you ran it in ("local scope), so the server would vanish as soon as you open Claude Code in exttt extasciitilde/my_workspace/session_N. exttt--scope user registers it for your whole user account, so it is available in every folder.

Then verify:

```
claude mcp list
```

You should see `github` in the list with a ✓ `Connected` status. **Now delete the scratch file containing your token.**

4c. Confirm it works

Open Claude Code in any directory and ask:

```
Use the GitHub MCP to list the five most recent public repos owned by my account.
```

If it returns a list, you are done. If it asks you to authenticate or returns an error, see Troubleshooting at the end.

Step 5: Install the Zotero MCP

The Zotero MCP lets Claude Code search your Zotero library, pull BibTeX, and insert citations into a draft. We use it for a citations demo and you will use it for your own papers afterward.

5a. Set up Zotero and import the practice library

- If you don't have a Zotero account, create a free one at **https://www.zotero.org/user/register**
- Install the **Zotero desktop app** from **https://www.zotero.org/download/** if you don't already have it — the MCP needs the desktop app running to read your library (including full-text PDFs)
- Open the desktop app and sign in

Now import the practice library. I have prepared a bundle of 9 journal articles (with 10 PDF attachments — one article has two PDFs) for the in-class demo. It lives in the course repo at:

```
session_4/docs/claude_code_zotero/
```

After running `git pull` from Step 1, that folder contains a `claude_code_zotero.rdf` file and a `files/` subfolder of PDFs.

To import:

- In the Zotero desktop app, click **File** → **Import...**
- Choose **"A file (BibTeX, RIS, Zotero RDF, etc.)"**, click **Continue**
- Navigate to `session_4/docs/claude_code_zotero/` and select `claude_code_zotero.rdf`
- On the next screen, check **"Place imported collections and items into new collection"** and **"Copy files to the Zotero storage folder"**
- Click **Continue**

You should end up with a new collection called `claude_code_zotero` in the left sidebar containing 9 papers (top-level items), with 10 PDF attachments in total — one paper has two PDFs. **Confirm by clicking on the collection and verifying you see 9 papers, each with an attachment paperclip icon, and that expanding them shows 10 PDFs in total.**

Critical: keep the `files/` folder *next to* the `.rdf` on disk during import — if you move just the .rdf or import it from elsewhere, Zotero will import the metadata but not the PDFs.

5b. Let Claude Code talk to your Zotero app

Zotero blocks other applications from connecting by default. Turn that off:

- Open the Zotero desktop app
- **Mac:** **Zotero** → **Settings** **Windows:** **Edit** → **Settings**
- Go to the **Advanced** tab
- Under **General**, check the box **"Allow other applications on this computer to communicate with Zotero"**
- Close the Settings window. **Leave Zotero running** — the MCP needs it open to query your library

The MCP also falls back to the Zotero Web API if the desktop app is closed, but the local connection is faster and gives Claude Code access to your PDFs — worth turning on.

5c. Get your Zotero API key and Library ID

- Go to **https://www.zotero.org/settings/keys**
- Your **Library ID** (a number, e.g. `1234567`) is shown at the top under "Your userID for use in API calls"
- Click **Create new private key**

- **Key name:** `claude-code-workshop`
- Check **Allow library access** (read-only is fine; check write access too if you want Claude to add items)
- Click **Save Key**, then **copy the key immediately**

Save both the key and your Library ID into a temporary scratch file.

5d. Install the `zotero-mcp-server` package

Mac:

```
pip install zotero-mcp-server
```

If `pip` is not found, use:

```
pip3 install zotero-mcp-server
```

Windows:

```
py -m pip install zotero-mcp-server
```

Make sure you type `zotero-mcp-server` (not `zotero-mcp`) in the install command: a different, unrelated package on PyPI is called `zotero-mcp`. The package we use installs a command-line program called `zotero-mcp`.

Then confirm it installed:

```
zotero-mcp --help
```

Expected: a help message listing options. If you see "command not found," close and reopen your terminal first.

5e. Add the Zotero MCP to Claude Code

Run (substituting your real values from Step 5c):

```
claude mcp add --scope user zotero -e ZOTERO_LOCAL=true -e ZOTERO_LIBRARY_ID=YOUR_LIBRARY_ID -e
ZOTERO_API_KEY=YOUR_API_KEY -e ZOTERO_LIBRARY_TYPE=user - zotero-mcp serve
```

Then verify:

```
claude mcp list
```

You should now see both `github` and `zotero` listed with ✓ `Connected`. **Delete the scratch file with your key and Library ID.**

`ZOTERO_LOCAL=true` tells the MCP to read from the Zotero desktop app (the setting you turned on in Step 5b), which is what gives Claude Code access to your PDFs. The API key and Library ID are used for the Zotero web API, which the MCP falls back to if the desktop app is not running. **The web API only sees the practice collection once the Zotero desktop app has finished syncing it to your online library**, so after the import, leave Zotero open until the sync icon at the top right stops spinning. As in Step 4, `--scope user` is what makes the server available in every folder.

5f. Confirm it works

Open Claude Code in any directory and ask:

```
Use the Zotero MCP to list 5 items from the claude_code_zotero collection.
```

If it returns titles from the practice library (Alt et al., Bisgaard, Hicks et al., etc.), you are done. **Make sure the Zotero desktop app is open** when you run this — if it isn't, the MCP will either fail or fall back to the (slower) Web API.

Step 6: Re-Read the Session 3 Git Cycle (5 min)

Session 4 leans heavily on the Git workflow from Session 3. Spend five minutes re-reading the slides from Session 3 (`slides/session3_slides.pdf` in the course repo), specifically the `commit` → `instruct` → `diff` → `decide` cycle and how branches work. The collaborative story exercise will move faster than this if those concepts are fresh.

Verification: Confirm Everything Works

Open a terminal and run each of the following. **Take a screenshot of the full output and send it to me** along with your GitHub username from Step 2.

```
claude --version
```

Expected: `2.1.x` or higher.

```
claude mcp list
```

Expected: both `github` and `zotero` listed with ✓ `Connected`.

```
zotero-mcp --help
```

Expected: a help message (any output is fine; "command not found" is the failure mode).

Also confirm:

- `git pull` on the course repo succeeds and you see a `session_4/` folder with new content
- The `claude_code_zotero` collection in your Zotero app contains 9 papers with 10 PDF attachments in total (one paper has two PDFs)
- Asking Claude Code "list 5 items from the claude_code_zotero collection" returns titles
- Asking Claude Code "list my five most recent GitHub repos" returns repo names
- You have emailed me your GitHub username

Troubleshooting

`claude mcp add` fails with "unknown flag": You are on an older Claude Code. Run `claude update` (Step 3) and try again.

GitHub MCP shows "Connected" but tool calls fail with 401/403: Your PAT is missing a permission. Regenerate it (Step 4a) and make sure *Contents*, *Issues*, and *Pull requests* are all

set to *Read and write*. Then re-run the `claude mcp add` command — it will overwrite the existing entry.

Zotero MCP shows "Connected" but returns empty results: Your library is genuinely empty, or `ZOTERO_LIBRARY_TYPE` is wrong. If you set up a *group* library instead of a personal one, change `user` to `group` in the env var.

`zotero-mcp: command not found` after `pip install`: Close and reopen your terminal. If still failing, the install went to a directory not on your PATH — email me with the output of `pip show zotero-mcp-server` and I will help.

`claude mcp list` shows `github` or `zotero` in one folder but not in another: The server was registered with the default "local" scope, which only applies to the folder where you ran `claude mcp add`. Re-run the command with `--scope user` (as shown above). If Claude Code says the server already exists, first remove the old entry with `claude mcp remove github` (or `claude mcp remove zotero`) and then run the `claude mcp add` command again.

You accidentally pasted a token into a chat or committed it to git: Revoke it immediately (GitHub: **https://github.com/settings/tokens**; Zotero: **https://www.zotero.org/settings/keys**) and generate a new one. Tokens are easy to rotate; a leaked token in a public repo is not.

Anything else: Email me at brady.allardice@upf.edu. Do not spend more than 10 minutes troubleshooting on your own. I would much rather help you before the workshop than lose time during it.